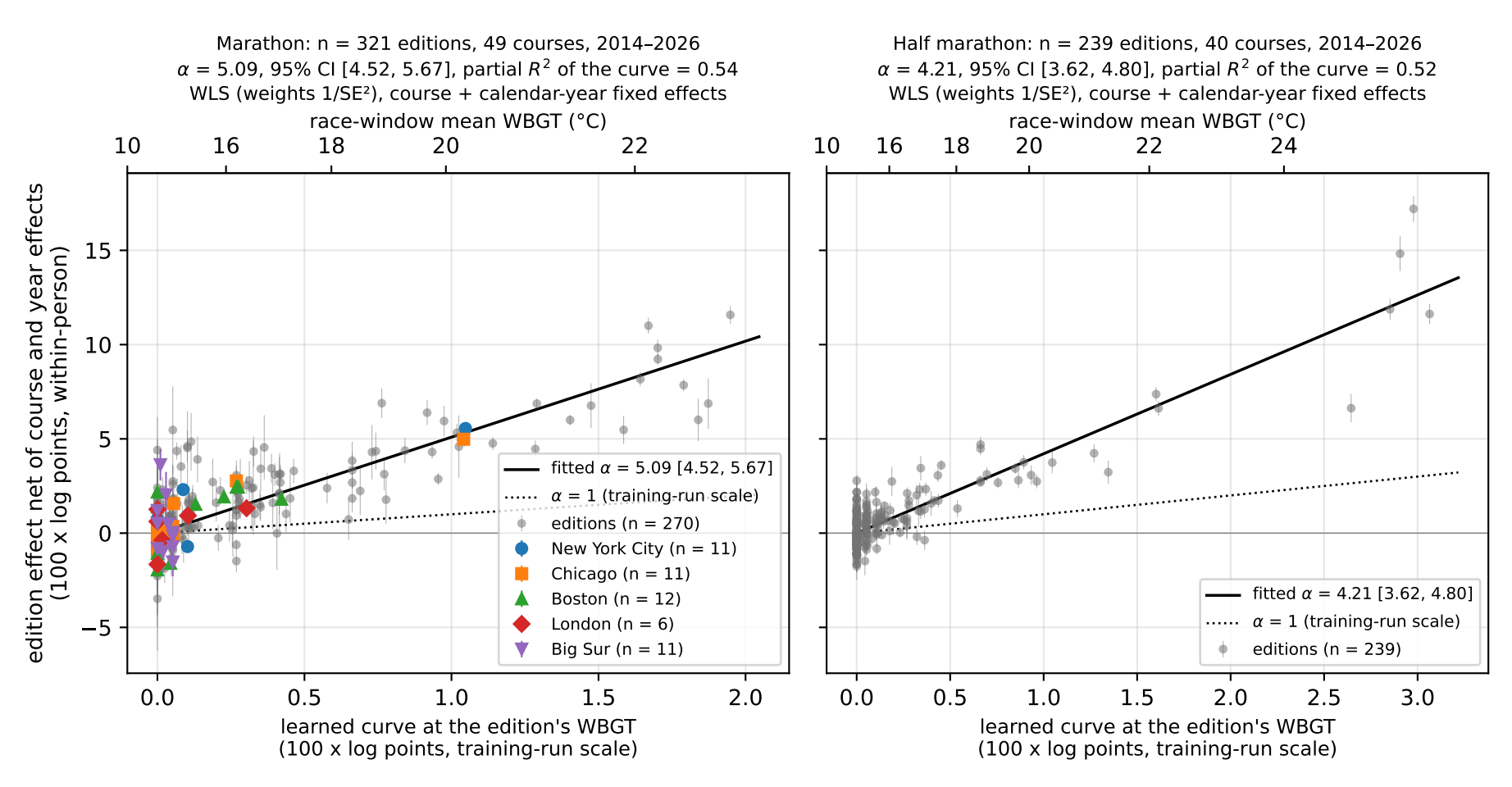
Marathon: n = 321 editions, 49 courses, 2014–2026
α = 5.09, 95% CI [4.52, 5.67], partial R² of the curve = 0.54
WLS (weights 1/SE²), course + calendar-year fixed effects
race-window mean WBGT (°C)
10
16
18
20
22
edition effect net of course and year effects
(100 x log points, within-person)
15
10
5
0
−5
fitted α = 5.09 [4.52, 5.67]
α = 1 (training-run scale)
editions (n = 270)
New York City (n = 11)
Chicago (n = 11)
Boston (n = 12)
London (n = 6)
Big Sur (n = 11)
0.0
0.5
1.0
1.5
2.0
learned curve at the edition's WBGT
(100 x log points, training-run scale)
Half marathon: n = 239 editions, 40 courses, 2014–2026
α = 4.21, 95% CI [3.62, 4.80], partial R² of the curve = 0.52
WLS (weights 1/SE²), course + calendar-year fixed effects
race-window mean WBGT (°C)
10
16
18
20
22
24
fitted α = 4.21 [3.62, 4.80]
α = 1 (training-run scale)
editions (n = 239)
0.0
0.5
1.0
1.5
2.0
2.5
3.0
learned curve at the edition's WBGT
(100 x log points, training-run scale)